SURVEYING & LEVELLING · NOTE SL15

Land Grading Survey and Land Levelling Design — the Plane Method

Course topics: IDE5.5 L9, IDE-503 T5

Land grading

Land grading reshapes the field surface to a planned grade to control the flow of water, check erosion and provide surface drainage, meeting the needs of irrigation and drainage. **Land levelling, land grading and land shaping** mean the same thing.

- Fields are graded to a **gentle uniform slope**, not to a truly level surface.
- Grading spreads irrigation water uniformly, conserves moisture, reduces erosion and gives surface drainage; in low-rainfall areas it raises infiltration by cutting runoff.

Grading criteria depend on: soil profile, existing land slope, rainfall, cropping pattern, method of irrigation, special features of the site and the farmer's preferences.

Phase	Work done
1. Rough grading	Remove abrupt irregularities — mounds, dunes, ridges; fill pits, depressions and gullies
2. Land levelling	Reshape the surface to the planned grade; moves large quantities of earth over long distances and leaves an irregular surface where loads are dumped
3. Land smoothing	Final pass that removes the remaining irregularities and leaves a plane surface

Plan the layout before the levelling

- Plan field boundaries, the irrigation distribution system, drains and farm roads first. The levelling plan must provide for supplying or absorbing excavated earth and keep a proper ratio between excavation and embankment.
- The irrigation system must deliver the required water to every part of the field at the required rate, without soil damage or excessive loss, be easy to reach and operate, and not obstruct other farm work.
- The topographic map decides the field layout, water conveyance, field drains, the irrigation method, the number and type of control structures and whether levelling is needed.

Element	Layout rules
Field channels	Along the upper reaches of the plots, generally perpendicular to the direction of irrigation; water surface 20–30 cm above the ground to be irrigated; nearly level (slope below 0.1%); locate by a route-type survey and take a centre-line profile for cut and fill
Fields	Workable size; ideal shape rectangular; access to every area; room for machines; easy to irrigate
Field roads	Above the irrigation channels and below the field drains
Drainage	Good surface and subsurface drainage is essential: remove excess rain promptly and safely; surface drainage helps prevent saline-alkali conditions; add artificial drains where natural drainage is poor; use interceptor drains at the upper edge of low-lying land against seepage from canals and over-irrigated areas. Plan irrigation and drainage together

Survey and staking

1. Run a grid survey — usually 25 m × 25 m; also 30 × 30 m, 20 × 20 m or 15 × 15 m.
2. Stake the grid points and take the first reading on a bench mark — known, or assumed as 0.00, 10.00 or 100.00 m.
3. Compute the RLs and draw contours at an interval suited to the slope.

Design methods

Four methods exist: the **plane method**, profile method, plan inspection method and contour adjustment method. The plane method is the most commonly used.

Plane method — step by step

1. **Locate the centroid of the field.** Rectangle: intersection of the diagonals. Triangle: intersection of the lines from the corners to the midpoints of the opposite sides. Irregular field: split it into rectangles and right-angled triangles and take moments about two perpendicular reference lines.
2. **Find the average elevation** — the elevation of the centroid:
3. **Fix the plane:** any plane through the centroid at this elevation gives equal volumes of cut and fill. With the chosen down-field and cross slopes, compute the formation level of every grid point.
4. **Compute cut and fill** at each point by comparing original and formation levels: **+ = fill, − = cut**. Check that ΣCut = ΣFill.

$$\text{Elevation of centroid} = \Sigma\ (\text{elevations of grid points}) \ /\ \text{number of grid points}$$

> Six grid points read 100.40, 100.25, 100.10, 100.30, 100.15 and 100.00 m. Average elevation = 601.20/6 = 100.20 m — the formation level at the centroid.

Cut–fill ratio

- Soil shrinks on compaction, so design more cut than fill: the cut–fill ratio is greater than 1, usually 1.2 to 1.6, and may range from 1.1 to 2.0 with soil type, quantity of earthwork and type of equipment.
- Make fill heights 10% more to allow for settlement.

Today most large-scale levelling is done with laser-guided land levellers, which control the scraper blade automatically to the design plane; the survey and plane-method design above still set the target grade.

Sources: ICAR eCourse 'Irrigation Engineering' — Dr K. N. Tiwari and Dr N. S. Raghuwanshi (IIT Kharagpur), reviewed by Dr S. D. Gorantiwar (MPKV Rahuri), Lessons 17–18 (land levelling design). Further reading: A. M. Michael and T. P. Ojha, Principles of

Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The most commonly used method of land levelling design is the

(a) profile method
(b) plane method
(c) plan inspection method
(d) contour adjustment method

Q2. Any plane through the centroid at the average elevation gives

(a) more cut
(b) more fill
(c) equal cut and fill
(d) zero slope

Q3. In land levelling computations, a + sign denotes

(a) cut
(b) fill
(c) error
(d) datum

Q4. The usual cut–fill ratio is

(a) 0.6–0.8
(b) 1.0
(c) 1.2–1.6
(d) 2.5–3.0

Q5. Field channel water surface should be above the field by

(a) 5–10 cm
(b) 20–30 cm
(c) 50–60 cm
(d) 1 m

Q6. The final operation of land levelling that leaves a plane surface is

(a) rough grading
(b) land smoothing
(c) sub-soiling
(d) bunding

Answers

Q1 — (b) plane method
Q2 — (c) equal cut and fill
Q3 — (b) fill
Q4 — (c) 1.2–1.6
Q5 — (b) 20–30 cm
Q6 — (b) land smoothing